

Recap: Riemannian Connections and Riemannian Hessian

Let $f: M \rightarrow \mathbb{R}$ $c: \mathbb{R} \rightarrow M$ $c(0) = x$, $c'(0) = v$.

$$g = f \circ c: \mathbb{R} \rightarrow \mathbb{R}$$

Using Taylor series expansion on g

$$\begin{aligned} g(t) &= \underbrace{g(0)}_{= f(c(t))|_{t=0}} + t \underbrace{g'(0)}_{= \frac{d}{dt} g(t)|_{t=0}} + \frac{t^2}{2} \underbrace{g''(0)}_{??} + \dots \\ &= f(c(0)) \\ &= f(x) \\ &= D(f \circ c)(t)[c'(t)]|_{t=0} \\ &= \langle \text{grad } f(c(t)), c'(t) \rangle_{c(t)}|_{t=0} \\ &= \langle \text{grad } f(x), v \rangle_x \end{aligned}$$

$$g''(0) = \frac{d}{dt} g'(t) \Big|_{t=0} = \frac{d}{dt} \langle \text{grad } f(c(t)), c'(t) \rangle_{c(t)} \Big|_{t=0}$$

Even for Riemannian submanifold of $\mathbb{R}^d$,
 $\in T_x M$??

$$\frac{d}{dt} \langle \text{grad } f(c(t)), c'(t) \rangle = \left\langle \frac{d}{dt} \text{grad } f(c(t)), \underbrace{c'(t)}_{\in T_x M} \right\rangle$$

$$+ \left\langle \underbrace{\text{grad } f(c(t))}_{\in T_x M}, \underbrace{c''(t)}_{\in T_x M?} \right\rangle$$

Let $V: M \rightarrow TM$ be a smooth vector field.

$$DV(x): T_x M \rightarrow T_{V(x)} TM$$

$$\therefore DV(x)[u] = \left. \frac{d}{dt} V(c(t)) \right|_{t=0} \text{ where}$$

$c: \mathbb{R} \rightarrow M$ (any smooth curve) s.t. $c(0) = x, c'(0) = u$

$$\therefore \left. \frac{d}{dt} V(c(t)) \right|_{t=0} \in T_{V(x)} TM \text{ and may } \underline{\underline{\text{NOT}}}$$

lie in $T_x M$.

The "right way"

Connections: Differentiating vector fields

Let $\mathcal{X}(M)$ denote the set of smooth vector fields on M .

Def: A connection or covariant derivative on manifold M is a map ∇ defined as:

$$\nabla: TM \times \mathcal{X}(M) \rightarrow TM: (u, V) \rightarrow \nabla_u V$$
$$((x, u), V) \rightarrow \nabla_u V(x)$$

NOTATIONS

$$\nabla_u V \equiv \nabla_u V(x) \quad u \in T_x M, V(x) \in T_x M.$$

Also note that $\nabla_u V \in T_x M$

Properties satisfied by Connections

$$\forall a, b \in \mathbb{R}, \quad \forall u, w \in T_x M$$

$$\forall f: M \rightarrow \mathbb{R} \text{ (smooth)} \quad \forall U, V, W \in \mathcal{X}(M)$$

1. Linearity in 1st argument:

$$\nabla_{aU+bW} V = a \nabla_U V + b \nabla_W V$$

2. Linearity in 2nd argument:

$$\nabla_u (aV + bW) = a \nabla_u V + b \nabla_u W$$

3. Leibniz Rule:

$$\nabla_u (fV) = Df(u)[u] \cdot V(u) + f(u) \cdot \nabla_u V$$

0. Smoothness: $(\nabla_u V)(u) := \nabla_{u(u)} V$

defines a smooth vector field

- For $M = \mathbb{R}^d$, $\nabla_u V = DV(u)[u]$

- On $M = \text{Embedded submanifold of } E$,

$\nabla_u V(u) = \underbrace{\text{Proj}_x}_{\text{orthogonal projection}} (\underbrace{DJ(u)[u]}_{\text{smooth extension}})$ is a valid connection

We demanded two more properties from connection to get "special connections"

④ Symmetry or torsion-free property:

$$[U, V]f = (\nabla_V U - \nabla_U V)f$$

⑤ Compatibility with the metric:

$$U\langle V, W \rangle = \langle \nabla_U V, W \rangle + \langle V, \nabla_U W \rangle$$

Connections satisfying properties 0-5 exist and are unique: called Levi-Civita connection or Riemannian Connection.

NOTATIONS:

$$\textcircled{a} \langle U, V \rangle : M \rightarrow \mathbb{R} : (\langle U, V \rangle)(x) = \langle U(x), V(x) \rangle_x$$

$$\textcircled{b} Uf : M \rightarrow \mathbb{R} : (Uf)(x) = Df(x)[U(x)]$$

$$\therefore Uf = \langle \text{grad } f, U \rangle \text{ if metric exists}$$

• Uf is called the action of the vector field.

U on a f^u through derivation.

$$\textcircled{c} [U, V]f = U(Vf) - V(Uf)$$

The commutator of the action (Uf) is denoted

by Lie bracket.

Properties ④-⑤ ensure that the R. Hessian is symmetric.

$$\bar{f} : \mathbb{R}^d \rightarrow \mathbb{R}.$$

Hessian of $\bar{f}$: linear map $\text{Hess } \bar{f}(x) : \mathbb{R}^d \rightarrow \mathbb{R}^d$

$$\text{Hess } \bar{f}(x)[v] = D(\text{grad } \bar{f})(x)[v]$$

$$= \lim_{t \rightarrow 0} \frac{\text{grad } \bar{f}(x+tv) - \text{grad } \bar{f}(x)}{t}$$

$\text{Hess } \bar{f}(x)$ is self adjoint w.r.t. $\langle \cdot, \cdot \rangle$ on $\mathbb{R}^d$

$$\langle \text{Hess } \bar{f}(x)(u), v \rangle = \langle \text{Hess } \bar{f}(x)(v), u \rangle \\ \forall u, v \in T\mathbb{R}^d.$$

Riemannian Hessian of $f: M \rightarrow \mathbb{R}$.

we let $\text{Hess } f(x): T_x M \rightarrow T_x M$ denote the Riemannian Hessian of f .

1) Riemannian Hessian is a linear map.

2) $\text{Hess } f(x)$ is the derivative of gradient vector field.

$$\therefore \text{Hess } f(x)(u) = \nabla_u \text{grad } f(x)$$

Thm: Given a Riemannian manifold M ,

Let ∇ be the R. connection, let

$\text{Hess } f(x)(u) = \nabla_u \text{grad } f(x)$, where

$u \in T_x M$ and $f: M \rightarrow \mathbb{R}$ (smooth). Then,

$$\langle \text{Hess } f(x)(u), v \rangle_x = \langle \text{Hess } f(x)(v), u \rangle_x \\ \forall u, v \in T_x M.$$

For Euclidean manifold, $\nabla_u V = DV(x)(u)$

is L.C. connection with any metric

For R. submanifold of E ,

$\nabla_u V = \text{Proj}_x (DV(x)(u))$ is the L.C. connection:

(Recap Ends)

Ex $f: M \rightarrow \mathbb{R}$ with $M = S^{d-1}$, $\mathbb{R}$ -submanifold

let $\bar{f}$ be a smooth extension of f . Compute $\text{Hess } f(x)$ in terms of $\bar{f}(x)$?

$$\text{grad } f(x) = \text{Proj}_x(\text{grad } \bar{f}(x))$$

$$= (I - xx^T) \text{grad } \bar{f}(x) =: \bar{G}(x)$$

$$\bar{G}(x) : \mathbb{R}^d \rightarrow \mathbb{R}^d$$

$$\begin{aligned} D\bar{G}(x)[u] &= (I - xx^T) D(\text{grad } \bar{f})(x)[u] \\ &\quad + (-ux^T) \text{grad } \bar{f}(x) \\ &\quad + (-x u^T) \text{grad } \bar{f}(x) \end{aligned}$$

$$\begin{aligned} \text{Hess } f(x)[u] &= \nabla_u \text{grad } f(x) \text{ where } \nabla \text{ is } \mathbb{R} \text{ connection} \\ &= \text{Proj}_x(D \text{grad } f(x)[u]) \\ &\quad \text{if } M \text{ is } \mathbb{R} \text{ submanifold of } \mathbb{R}^d \end{aligned}$$

$$= \text{Proj}_x(D\bar{G}(x)[u])$$

$$= (I - \alpha \alpha^T) \underbrace{\nabla \text{grad } \bar{f}(n) [u]}_{\text{Hess } \bar{f}(n) [u]} - \alpha^T \text{grad } \bar{f}(n) u$$

$$= \text{Proj}_n (\text{Hess } \bar{f}(n) [u]) - \underbrace{\alpha^T \text{grad } \bar{f}(n) u}_{\text{a correction term.}}$$

It is not enough to Project $\text{Hess } \bar{f}(n) [u]$ due to the presence of the correction term.

Differentiating vector fields along curves

We wanted to get 2nd order Taylor series expansion of $g(t) = f(c(t))$

$$g(t) = g(0) + t g'(0) + \frac{t^2}{2} g''(0) + O(t^3)$$

$$g'(t) = Df(c(t))[c'(t)] = \langle \text{grad}(f \circ c)(t), c'(t) \rangle_{c(t)}$$

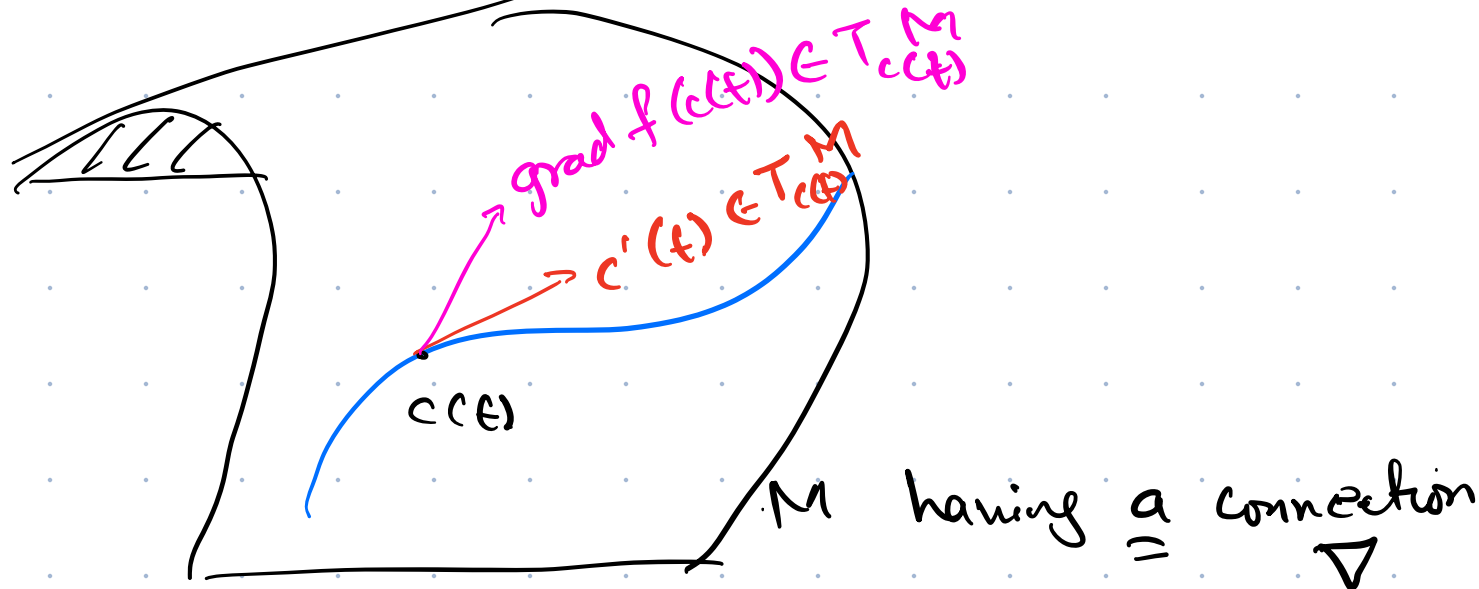
$$g''(0) = \frac{d}{dt} g'(t) \Big|_{t=0} = ??$$

The connection ∇ that we defined earlier works on a vector field of M

$c'(t)$ is not a vector field on M .

$c'(t)$ is a vector field on $\overset{\text{curve}}{c(t) \in M}$

Similarly $\text{grad}(f \circ c)(t)$ is a vector field on $\text{curve } c \in M$



Def: A smooth vector field along curve c is a

smooth map $Z: \mathbb{R} \rightarrow TM$ s.t. $Z(t) \in T_{c(t)}M \forall t$

The set of smooth vector fields along ' c ' is denoted by $\mathcal{X}(c)$

As soon as we have a connection ∇ (need not be R. connection), $\exists$ a unique operator

$\frac{D}{dt}: \mathcal{X}(c) \rightarrow \mathcal{X}(c)$ satisfying the following

three properties: $\forall a, b \in \mathbb{R}, \gamma, z \in \mathcal{X}(c), g: \mathbb{R} \rightarrow \mathbb{R}$
Smooth

1. Linearity: $\frac{D}{dt} (a\gamma + bz)(t) = a \frac{D}{dt} \gamma(t) + b \frac{D}{dt} z(t)$

2. Leibniz's rule:

$$\frac{D}{dt} (gZ)(t) = g'(t) \cdot Z(t) + g(t) \cdot \frac{D}{dt} Z(t)$$

3. Chain Rule:

$$\frac{D}{dt} U(c(t)) = \nabla_{c'(t)} U(c(t)) \quad \forall U \in \mathcal{X}(M)$$

If ∇ is compatible with the metric, then

$\frac{D}{dt}$ also satisfies product rule:

$$\begin{aligned} \frac{d}{dt} \langle \gamma(t), z(t) \rangle_{c(t)} &= \left\langle \frac{D}{dt} \gamma(t), z(t) \right\rangle_{c(t)} + \\ &\quad \left\langle \gamma(t), \frac{D}{dt} z(t) \right\rangle_{c(t)} \end{aligned}$$

We call $\frac{D}{dt}$ the covariant derivative induced

by ∇

We still have not discussed how to actually compute $\frac{D}{dt}$. For special cases:

In Euclidean Space:
$$\begin{aligned}\frac{D}{dt} Z(t) &= \frac{d}{dt} Z(t) \\ &= \lim_{\delta \rightarrow 0} \frac{Z(t+\delta) - Z(t)}{\delta}\end{aligned}$$

In R. submanifold of $\mathbb{R}^d$:

$$\frac{D}{dt} Z(t) = \underset{\uparrow}{\text{Proj}}_{c(t)} \left(\frac{d}{dt} Z(t) \right)$$

orthonormal w.r.t. Euclidean metric.

Acceleration and geodesics:

Let M be a manifold with connection ∇ and induced $\frac{D}{dt}$

The (intrinsic) acceleration along c is

$$c''(t) = \frac{D}{dt} c'(t)$$

(depends on the choice of ∇)

A geodesic is a curve 'c' s.t. $c''(t) = 0 \forall t$.

Geodesics generalizes the notion of straight lines to non-linear spaces

Ex $M = S^{d-1}$ and $\mathbb{R}$ submanifold of $\mathbb{R}^d$.

$$c(t) = \cos(t)u + \sin(t)v.$$

Prove that $c''(0) = 0$

Taylor series - second order:

M : $\mathbb{R}$. manifold with $\mathbb{R}$. connection ∇ and associated $\frac{D}{dt}$.

$$f: M \rightarrow \mathbb{R}, c: \mathbb{R} \rightarrow M, c(0) = x, c'(0) = v$$

$$g(t) = g(0) + t g'(0) + \frac{t^2}{2} g''(0) + O(t^3)$$

$$g'(t) = \langle \text{grad } f(c(t)), c'(t) \rangle_{c(t)}$$

$$g''(t) = \frac{d}{dt} g'(t)$$

$$\stackrel{\text{by product rule}}{=} \left\langle \frac{D}{dt} \text{grad}(f \circ c)(t), c'(t) \right\rangle_{c(t)}$$

$$+ \left\langle \text{grad}(f \circ c)(t), \frac{D}{dt} c'(t) \right\rangle_{c(t)}$$

$$\stackrel{\text{chain rule}}{=} \left\langle \nabla_{c'(t)} \text{grad}(f \circ c)(t), c'(t) \right\rangle_{c(t)}$$

$$+ \left\langle \text{grad}(f \circ c)(t), c''(t) \right\rangle_{c(t)}$$

$$= \langle \text{Hess } f(x) [c'(t)], c'(t) \rangle_{c(t)}$$

$$+ \langle \text{grad } f(c(t)), c''(t) \rangle_{c(t)}$$

$$\therefore g''(0) = \langle \text{Hess } f(x) [v], v \rangle_x + \langle \text{grad } f(x), c''(0) \rangle_x$$

$$f(c(t)) = f(x) + t \langle \text{grad } f(x), v \rangle_x$$

$$+ \frac{t^2}{2} \left[\langle \text{Hess } f(x) [v], v \rangle_x + \langle \text{grad } f(x), w \rangle_x \right]$$

$$+ O(t^3)$$

$$\text{where } c(0) = x, \quad c'(0) = v, \quad c''(0) = w.$$

When will this term $\langle \text{grad } f(x), c''(0) \rangle_x = 0$??

① $\text{grad } f(x) = 0$ (critical point)

② $c''(0) = 0$

In optimization, the curves we work with are retractions. So we can define

Special retraction curves called "**Second Order Retraction**" which have zero

initial acceleration, i.e. $\forall (x, v) \in TM,$

$c(t) = R_x(tv)$ satisfies $c(0) = x$
 $c'(0) = v$
 $c''(0) = 0$